

INTRODUCTION TO HEALTH ECONOMICS

Quiz 2 — Detailed Answer Key

The individual's utility function is $U(I) = 1000I - I^2$.

Baseline incomes: healthy = 1000, sick = 0.

Probability of falling sick: $p = 0.5$.

Step 1: Expected Utility Without Insurance

$$EU(\text{no ins}) = 0.5 \times U(1000) + 0.5 \times U(0)$$

$$U(1000) = 1000(1000) - (1000)^2 = 1,000,000 - 1,000,000 = 0$$

$$U(0) = 0$$

$$EU(\text{no ins}) = 0.5(0) + 0.5(0) = 0$$

Step 2: Expected Utility Under Contract A (Full Insurance, Premium = 600)

Full insurance covers the entire income loss. With premium 600, net income = $1000 - 600 = 400$ in both states (certain outcome).

$$EU_a = U(400) = 1000(400) - (400)^2 = 400,000 - 160,000 = 240,000$$

Step 3: Expected Utility Under Contract B (Fair Partial Insurance, Payout = q , Premium = $0.5q$)

Under Contract B:

- If healthy: income = $1000 - 0.5q$ (pays premium, no payout)
- If sick: income = $0 - 0.5q + q = 0.5q$ (pays premium, receives payout q)

$$EU^B = 0.5 \times U(1000 - 0.5q) + 0.5 \times U(0.5q)$$

Expanding $U(1000 - 0.5q)$:

$$\begin{aligned} &= 1000(1000 - 0.5q) - (1000 - 0.5q)^2 \\ &= 1,000,000 - 500q - (1,000,000 - 1000q + 0.25q^2) \\ &= 500q - 0.25q^2 \end{aligned}$$

Expanding $U(0.5q)$:

$$= 1000(0.5q) - (0.5q)^2 = 500q - 0.25q^2$$

Both terms are identical, so:

$$EU^B = 0.5(500q - 0.25q^2) + 0.5(500q - 0.25q^2) = 500q - 0.25q^2$$

Step 4: Solve for Indifference ($EU_a = EU^B$)

$$240,000 = 500q - 0.25q^2$$

$$0.25q^2 - 500q + 240,000 = 0$$

Multiplying through by 4:

$$q^2 - 2000q + 960,000 = 0$$

$$q = 1200 \quad \text{or} \quad q = 800$$

Step 5: Select the Economically Relevant Root

- **q = 1200:** Payout exceeds healthy income (1000). This implies overinsurance — the sick-state payout exceeds the healthy baseline. Not feasible in a standard model. Discard.
- **q = 800:** Premium = $0.5 \times 800 = 400$. Healthy net income = $1000 - 400 = 600$. Sick net income = $-400 + 800 = 400$. Both within $[0, 1000]$. Economically valid.

► **Answer: q = 800**

Step 6: Economic Interpretation

Why indifference at partial coverage? Contract A eliminates all risk but charges 600. Contract A's premium (600) exceeds the actuarially fair full-coverage premium ($0.5 \times 1000 = 500$) by 100. This loading leads the individual to prefer partial insurance if priced generously enough. At $q = 800$, the fair premium of 400 is low enough to compensate for the residual income variance, making her exactly indifferent.

Marks Breakdown

Component	Marks
Correct EU without insurance / setup	1
Correct EU under Contract A	2
Correct EU under Contract B	3
Solving the quadratic correctly	2
Selecting the right root with justification	1
Economic interpretation	1
Total	10